torch.from_dlpack#

https://pytorch.org/docs/stable/generated/torch.from_dlpack.html

Description:

Converts a tensor from an external library into a torch.Tensor. The returned PyTorch tensor will share the memory with the input tensor (which may have come from another library). Note that in-place operations will therefore also affect the data of the input tensor. This may lead to unexpected issues (e.g., other libraries may have read-only flags or immutable data structures), so the user should only do this if they know for sure that this is fine. ext_tensor (object with __dlpack__ attribute, or a DLPack capsule) – The tensor or DLPack capsule to convert. If ext_tensor is a tensor (or ndarray) object, it must support the __dlpack__ protocol (i.e., have a ext_tensor.__dlpack__ method). Otherwise ext_tensor may be a DLPack capsule, which is an opaque PyCapsule instance, typically produced by a to_dlpack function or method.

Function Signature

```
torch.from_dlpack(ext_tensor) → Tensor[source]#

Parameters

Return type
```

Returns:

Tensor

Code Examples

```
>>> import torch.utils.dlpack
>>> t = torch.arange(4)

# Convert a tensor directly (supported in PyTorch >= 1.10)
>>> t2 = torch.from_dlpack(t)
>>> t2[:2] = -1  # show that memory is shared
>>> t2
tensor([-1, -1,  2,  3])
>>> t
tensor([-1, -1,  2,  3])

# The old-style DLPack usage, with an intermediate capsule
object
>>> capsule = torch.utils.dlpack.to_dlpack(t)
>>> capsule
<capsule object "dltensor" at ...>
>>> t3 = torch.from_dlpack(capsule)
>>> t3
tensor([-1, -1,  2,  3])
>>> t3[0] = -9  # now we're sharing memory between 3 tensors
>>> t3
tensor([-9, -1,  2,  3])
>>> t2
tensor([-9, -1,  2,  3])
>>> t
tensor([-9, -1,  2,  3])
```

Detailed Documentation

Documentation

Converts a tensor from an external library into a torch.Tensor.

The returned PyTorch tensor will share the memory with the input tensor (which may have come from another library). Note that in-place operations will therefore also affect the data of the input tensor. This may lead to unexpected issues (e.g., other libraries may have read-only flags or immutable data structures), so the user should only do this if they know for sure that this is fine.

ext_tensor (object with __dlpack__ attribute, or a DLPack capsule) –

The tensor or DLPack capsule to convert.

If ext_tensor is a tensor (or ndarray) object, it must support the __dlpack__ protocol (i.e., have a ext_tensor.__dlpack__ method). Otherwise ext_tensor may be a DLPack capsule, which is an opaque PyCapsule instance, typically produced by a to_dlpack function or method.

device (torch.device or str or None) – An optional PyTorch device specifying where to place the new tensor. If None (default), the new tensor will be on the same device as ext_tensor.

copy (bool or None) – An optional boolean indicating whether or not to copy self. If None, PyTorch will copy only if necessary.